Online Appendix to "Anatomy of the Treasury Market"

Manav Chaudhary Julie Zhiyu Fu Haonan Zhou

Appendix A develops the estimation framework behind Section 3: time-varying elasticities, loadings, and sizes; common factors; omitted-factor bias; and structural interpretations. Appendix B describes the data and common factors, checks the robustness of identification, and reports additional results on multipliers, the yield decomposition, and the safe haven. Proofs, additional results, and further discussions are in the Supplementary Material (Appendices S1 to S4), available at
https://anatomyoftreasury.github.io/AnatomyOfTreasury/Treasury_GIV_supplement.pdf.

A Methodological Appendix

A.1 General Framework

Here we generalize the model of Section 2.2. Notation: for entities $i = 1, \ldots, N$ and periods $t = 1, \ldots, T$, let $q_{i,t} \in \mathbb{R}$ denote the quantity of entity i at time t, $p_t \in \mathbb{R}$ the price at time t, and $S_{i,t} > 0$ the (exogenous) size of entity i at time t. Let $\mathbf{C}_{i,t} \in \mathbb{R}^{N_\zeta}$ parameterize elasticities and $\mathbf{X}_{i,t} \in \mathbb{R}^{N_\beta}$ collect exogenous controls. Let $\boldsymbol{\zeta} \in \mathbb{R}^{N_\zeta}$ denote the elasticity vector and $\boldsymbol{\beta} \in \mathbb{R}^{N_\beta}$ denote the loading vector. We use the subscript S to denote size-weighted sums (e.g., $\mathbf{C}_{S,t} := \sum_i S_{i,t}\, \mathbf{C}_{i,t}$). The general model is specified as follows:

$$\left.\begin{array}{rl} q_{i,t} & = -p_t \times \mathbf{C}_{i,t}'\boldsymbol{\zeta} + \mathbf{X}_{i,t}'\boldsymbol{\beta} + u_{i,t} \\ 0 & = \sum_i S_{i,t} q_{i,t} \end{array}\right\} \implies p_t = \frac{1}{\mathbf{C}_{S,t}'\boldsymbol{\zeta}} \left[\mathbf{X}_{S,t}'\boldsymbol{\beta} + u_{S,t}\right], \tag{A.1}$$

which allows for the following flexibility:

- The elasticity is parameterized by the matrix $\mathbf{C}_{i,t}$, which can be entity-specific and time-varying;
- The control variable $\mathbf{X}_{i,t}$ can also be entity&time-specific;

- The size vector $S_{i,t}$ is also allowed to be time-varying; we do not require it to sum to 1. The only assumption we impose on sizes is positivity, $S_{i,t} > 0$. This is not a restriction as one can always simply swap the sign of $q_{i,t}$ and $S_{i,t}$ to satisfy the assumption. For example, the supply can be specified as a sector with size 1 and a higher supply enters as a negative $q_{i,t}$, so that its elasticity has the same sign as other sectors.

The simple model can be recovered from the general specification by setting $\mathbf{C}_{i,t}$ to be the entity fixed effects, and $\mathbf{X}_{i,t}$ to be the interaction of entity fixed effects with the aggregate factors $\boldsymbol{\eta}_t$.

The moment conditions used to identify the elasticity vector $\boldsymbol{\zeta}$ remain the same as before, only it is now conditional on all other exogenous variables $(\mathbf{X}_t, \mathbf{C}_t, \mathbf{S}_t)$:

$$\mathbb{E}\left[u_{i,t}u_{j,t} \mid \mathbf{X}_t, \mathbf{C}_t, \mathbf{S}_t\right] = 0 \text{ for all } i \neq j. \tag{A.2}$$

We discuss the handling of control variables $\mathbf{X}_{i,t}$ in detail in Appendix A.2. At the high level, when $\mathbf{X}_{i,t}$ are observable, we partial out the effect of controls by (i) regressing $q_{i,t}$ on $\mathbf{X}_{i,t}$ to obtain $q^{\varepsilon}_{i,t}$ and (ii) regressing p_t on $\mathbf{X}_{S,t} := \sum_i S_{i,t}\mathbf{X}_{i,t}$ to obtain p^{ε}_t. We then form the optimal granular instrumental variables (GIV) estimator for $\boldsymbol{\zeta}$ using the residuals $q^{\varepsilon}_{i,t}$ and p^{ε}_t. We provide the definition of the optimal GIV estimator and its asymptotic properties below, and provide the proof in Supplementary Appendix S4.1.

Definition A.1. Let $\boldsymbol{\psi}_t(\mathbf{z})$ be a vector of length $\frac{N(N-1)}{2}$ with subscript ij $(i \neq j)$ denote the row corresponding the entity pair (i, j) :

$$\psi_{ij,t}(\mathbf{z}) \equiv \hat{u}_{i,t}(\mathbf{z})\,\hat{u}_{j,t}(\mathbf{z}) \equiv \left(q^{\varepsilon}_{i,t} + p^{\varepsilon}_t \mathbf{C}'_{i,t}\mathbf{z}\right)\left(q^{\varepsilon}_{j,t} + p^{\varepsilon}_t \mathbf{C}'_{j,t}\mathbf{z}\right).$$

Denote $\mathbf{V}^{\psi} \equiv \mathbb{E}\left[\boldsymbol{\psi}_t(\boldsymbol{\zeta})\,\boldsymbol{\psi}_t(\boldsymbol{\zeta})'\right] = \operatorname{diag}\left(\sigma_1^2\sigma_2^2, \sigma_1^2\sigma_3^2, \ldots, \sigma_{N-1}^2\sigma_N^2\right)$. Denote $\hat{\mathbf{W}}^*_t(\mathbf{z})$ as weighting matrix of the size $\frac{N(N-1)}{2} \times N_{\zeta}$ such that

$$\hat{\mathbf{W}}^*_{ij,k,t}(\mathbf{z}) = \frac{1}{\mathbf{C}'_{S,t}\mathbf{z}}\left(\frac{S_{j,t}C_{it,k}}{\hat{\sigma}_i^2(\mathbf{z})} + \frac{S_{i,t}C_{jt,k}}{\hat{\sigma}_j^2(\mathbf{z})}\right), \tag{A.3}$$

where $\hat{\sigma}_i^2(\mathbf{z}) \equiv \hat{\mathbb{E}}\left[\hat{u}_{it}(\mathbf{z})^2\right]$. The optimal GIV estimator $\hat{\boldsymbol{\zeta}}$ for the general model solves the following sample moment conditions:

$$\hat{\mathbb{E}}\left[\hat{\mathbf{W}}^*_t\left(\hat{\boldsymbol{\zeta}}\right)'\boldsymbol{\psi}_t\left(\hat{\boldsymbol{\zeta}}\right)\right] = \mathbf{0}. \tag{A.4}$$

While (A.3) looks complicated, it has a very intuitive interpretation.

Example 1. Consider the simple case described in the main text, where each investor has its own time-invariant elasticity and size, namely, $S_{jt} = S_j$ and $C_{it,k} = \begin{cases} 1 & i = k \\ 0 & i \neq k \end{cases}$. In this case, the optimal GIV estimator solves:

$$\frac{1}{T}\sum_t \hat{u}_{it}\left(\hat{\boldsymbol{\zeta}}\right) \underbrace{\sum_{j\neq i} S_j \hat{u}_{jt}\left(\hat{\boldsymbol{\zeta}}\right)}_{\hat{u}_{S(-i),t}\left(\hat{\boldsymbol{\zeta}}\right)} = 0 \quad \forall i.$$

That is, the optimal GIV estimator weights different sectors using their sizes—entities with larger influence are weighted more in the estimation.

Example 2. If we further assume homogeneous elasticity across all entities, the optimal GIV estimator further weights entities sharing the same elasticity using their own residual volatilities:

$$\frac{1}{T}\sum_t \sum_i \frac{1}{\sigma_i^2} \hat{u}_{it}\left(\hat{\boldsymbol{\zeta}}\right) \hat{u}_{S(-i),t}\left(\hat{\boldsymbol{\zeta}}\right) = 0.$$

This is akin to the generalized least squares (GLS) estimator: entities with noisier observations are weighted less in estimation.

Similar to the simple model, the optimal GIV estimator in this case is also asymptotically efficient. We formally state the assumptions needed for the optimal GIV estimator:
Assumption A.1 (Full model)**.**

1. The model is specified as in (A.1), with $\mathbf{S}_t > 0$ and $\mathbf{C}'_{S,t}\boldsymbol{\zeta} \neq 0$ with probability 1.
2. For each entity i, $u_{i,t}$ is i.i.d. with $\mathbb{E}\left[u_{i,t} \mid \mathrm{C}_t, \mathrm{S}_t, \mathbf{X}_t\right] = 0$ and $\mathbb{E}\left[u_{i,t}^2 \mid \mathrm{C}_t, \mathrm{S}_t, \mathbf{X}_t\right] = \sigma_i^2 > 0$.[1]
3. For any $i \neq j$, $u_{i,t}$ is independent of $u_{j,t}$ conditional on $(\mathrm{C}_t, \mathrm{S}_t, \mathbf{X}_t)$.
4. $\mathbb{E}\left[\mathbf{X}_{i,t}\mathbf{X}'_{i,t}\right]$ are non-singular.
5. The parameter space for $(\boldsymbol{\zeta}, \boldsymbol{\beta})$ is compact and the true parameter is in the interior of the parameter space.

To ensure the estimator is well-behaved, we make one additional assumption on the

[1]This i.i.d. condition is used for efficiency. Consistency of $\hat{\boldsymbol{\zeta}}$ only requires $\mathbb{E}[u_{i,t} \mid \mathbf{C}_t, \mathbf{S}_t, \mathbf{X}_t] = 0$ and finite second moments. If $u_{i,t}$ exhibits serial correlation, the optimal (efficient) weighting replaces σ_i^2 by the long-run variance $\Omega_i := \sum_{\ell=-\infty}^{\infty} \gamma_i(\ell)$ with $\gamma_i(\ell) = \mathbb{E}[u_{i,t}u_{i,t-\ell}]$ (i.e., HAC/Newey–West). The resulting optimal weight retains the same form with σ_i^2 replaced by Ω_i.

weighting matrix, where $\mathbf{W}_t \equiv \mathbf{W}_t(\boldsymbol{\zeta})$ denotes a weighting matrix evaluated at the true parameter.

Assumption A.2. *$\mathbb{E}[\mathbf{W}_t'\mathbf{W}_t]$ is non-singular. $\mathbb{E}[\mathbf{W}_t'\boldsymbol{\psi}_t(\mathbf{z})] = 0$ if and only if $\mathbf{z} = \boldsymbol{\zeta}$.*[2]

Theorem A.1 (Asymptotic efficiency in the general model)**.** *Given the moment conditions (A.2), under Assumptions A.1 and A.2, the optimal GIV estimator $\hat{\boldsymbol{\zeta}}$ is consistent and asymptotically normal:*

$$\sqrt{T}\left(\hat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}\right) \xrightarrow{d} \mathcal{N}\left(0, \mathbf{V}_\zeta^{W*}\right),$$

for $T \to \infty$, where

$$\mathbf{V}_\zeta^{W*} = \left(\mathbb{E}\left[\mathbf{W}_t^{*\prime}\mathbf{V}^{\psi}\mathbf{W}_t^{*}\right]\right)^{-1}. \tag{A.5}$$

Moreover, $\mathbf{V}_\zeta^{W*}$ achieves the semiparametric efficiency bound.

Proof. See Supplementary Appendix S4.1. □

Standard errors of nonlinear functions of the estimates, such as the macro multiplier, price responses, and total loadings, follow from the delta method. Leave-one-out estimates drop a sector's moment conditions, which breaks the market clearing behind (A.5); they use the sandwich variance, with $\mathbf{D}_t \equiv \partial\boldsymbol{\psi}_t(\boldsymbol{\zeta})/\partial\boldsymbol{\zeta}'$:

$$\mathbf{V}_\zeta^{W} = \mathbb{E}[\mathbf{W}_t'\mathbf{D}_t]^{-1}\,\mathbb{E}[\mathbf{W}_t'\boldsymbol{\psi}_t\boldsymbol{\psi}_t'\mathbf{W}_t]\,\mathbb{E}[\mathbf{D}_t'\mathbf{W}_t]^{-1}. \tag{A.6}$$

Bias-corrected OLS interpretation. In the simple case of the first example, the estimator can also be written as a bias-corrected OLS estimator. Market clearing implies $\hat{u}_{S(-i),t} = \hat{\zeta}_S p_t^\varepsilon - S_i\hat{u}_{i,t}$, where $\hat{\zeta}_S \equiv \sum_j S_j\hat{\zeta}_j$. Substituting into that example's moment condition gives

$$\hat{\zeta}_i = -\frac{\hat{\mathbb{E}}\left[q_{i,t}^\varepsilon p_t^\varepsilon\right]}{\hat{\mathbb{E}}\left[(p_t^\varepsilon)^2\right]} + \frac{S_i\hat{\sigma}_i^2}{\hat{\zeta}_S\,\hat{\mathbb{E}}\left[(p_t^\varepsilon)^2\right]}. \tag{A.7}$$

The first term is the OLS estimate from regressing quantities on prices. It is efficient for an infinitesimal sector but biased for a sector large enough to move the price: the sector's own demand shocks raise the price, so its demand appears less elastic than it is. The second term removes this bias using the sector's own price impact, and it vanishes when the sector is infinitesimal or the market is perfectly elastic. Equation (A.7) also gives an alternative, and sometimes more stable, algorithm. Supplementary Appendix S3.2 illustrates the correction for households and mutual funds.

[2]For the optimal weighting matrix $\mathbf{W}_t^*$ and non-negative $\mathbf{C}_{i,t}$, the second part follows from the first; Supplementary Appendix S2.4.3 gives the argument. With unrestricted $\mathbf{C}_{i,t}$ multiple roots can arise, and uniqueness must be verified case by case.

Comparison with the standard GIV. The optimal GIV differs from the standard GIV of Gabaix and Koijen (2024) in two respects. First, it accommodates heterogeneous elasticities and factor loadings through $\mathbf{C}_{i,t}$ and $\mathbf{X}_{i,t}$, without parameterizing them on observable characteristics. Second, it weights the moment conditions by both size and precision and is therefore efficient, whereas the standard GIV instrument, a size-weighted minus precision-weighted average of quantities, loses power when larger entities have less volatile shocks. In exchange, the optimal GIV must be solved numerically, and it does not remove unobserved common factors by cross-sectional demeaning. Supplementary Appendix S2.2 formalizes the comparison, and Supplementary Appendix S2.4 discusses missing sectors, measurement errors, and the uniqueness of the solution.

A.2 Common Factors and Identification of Factor Loadings $\boldsymbol{\lambda}_i$

We now consider external common factors. They can be directly observable, such as macrofinancial variables, or unobservable principal components extracted from granular data *outside of the model*. Supplementary Appendix S2.1 discusses the method to extract the unobserved common factors *within the model*.

With external common factors, we proceed with the following steps. We have an additional set of moment conditions to identify the coefficients on $\mathbf{X}_{i,t}$:

$$\mathbb{E}\left[\mathbf{X}_{i,t}u_{i,t}\right] = \mathbb{E}\left[\mathbf{X}_{i,t}(q_{i,t} + p_t\mathbf{C}'_{i,t}\boldsymbol{\zeta} - \mathbf{X}'_{i,t}\boldsymbol{\beta})\right] = \mathbf{0}. \tag{A.8}$$

Hence, in principle, $\boldsymbol{\zeta}$ and $\boldsymbol{\beta}$ are estimated jointly in a generalized method of moments (GMM) system with both sets of moment conditions. Algorithmically, for each guess of $\boldsymbol{\zeta}$, we can estimate $\boldsymbol{\beta}$ by regressing $q_{i,t} + p_t\mathbf{C}'_{i,t}\boldsymbol{\zeta}$ on $\mathbf{X}_{i,t}$; then we use the residuals to form the moment conditions in (A.2).

We can further simplify the algorithm by utilizing the linearity of the OLS estimator. Notice that regressing $q_{i,t} + p_t\mathbf{C}'_{i,t}\boldsymbol{\zeta}$ on $\mathbf{X}_{i,t}$ is equivalent to regressing $q_{i,t}$ on $\mathbf{X}_{i,t}$ and $p_t\mathbf{C}'_{i,t}$ on $\mathbf{X}_{i,t}$, and combine the residuals. Hence, we can follow the three-step algorithm below:

0. (If needed: Use principal component analysis (PCA) to extract unobserved common factors from external datasets, and include them in $\mathbf{X}_{i,t}$)
1. Regress $q_{i,t}$ and $p_t\mathbf{C}'_{i,t}$ on $\mathbf{X}_{i,t}$ to estimate $\hat{\boldsymbol{\beta}}^q$ and $\hat{\boldsymbol{\beta}}^{pC}$.
2. Take the residual from the first step, $q^{\varepsilon}_{i,t} \equiv q_{i,t} - \mathbf{X}'_{i,t}\hat{\boldsymbol{\beta}}^q$ and $\mathbf{pC}^{\varepsilon}_{i,t}{}' = p_t\mathbf{C}'_{it} - \mathbf{X}'_{i,t}\hat{\boldsymbol{\beta}}^{pC}$,

and form the moment conditions:

$$\psi_{ijt}^{\varepsilon}(\mathbf{z}) \equiv \hat{u}_{it}^{\varepsilon}(\mathbf{z})\,\hat{u}_{jt}^{\varepsilon}(\mathbf{z}) \equiv \left(q_{i,t}^{\varepsilon} + \mathbf{p}\mathbf{C}_{i,t}^{\varepsilon\prime}\mathbf{z}\right)\left(q_{j,t}^{\varepsilon} + \mathbf{p}\mathbf{C}_{j,t}^{\varepsilon\prime}\mathbf{z}\right).$$

Proceed with the optimal weighting matrix defined in (A.3) to estimate $\hat{\zeta}$.[3]

3. Form the estimator for $\boldsymbol{\beta}$ as:

$$\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{q} + \hat{\boldsymbol{\beta}}^{pC}\hat{\boldsymbol{\zeta}}.$$

The following proposition shows the asymptotic variance of $\hat{\boldsymbol{\zeta}}$ and $\hat{\boldsymbol{\beta}}$.

Proposition A.1. *The asymptotic variance formula for $\hat{\boldsymbol{\zeta}}$ in Theorem A.1 still holds with common factors. $\hat{\boldsymbol{\beta}}$ is a consistent estimator for $\boldsymbol{\beta}$, and its asymptotic variance for $\hat{\boldsymbol{\beta}}$ is given as:*

$$Var(\hat{\boldsymbol{\beta}}) = \frac{1}{T}\left[Var^{OLS}(\hat{\boldsymbol{\beta}}) + \boldsymbol{\beta}^{pC}Var(\hat{\boldsymbol{\zeta}})\boldsymbol{\beta}^{pC\prime}\right] \tag{A.9}$$

$$Var^{OLS}\left(\hat{\boldsymbol{\beta}}\right) = \left(\sum_i \mathbb{E}\left[\mathbf{X}_{i,t}\mathbf{X}_{i,t}'\right]\right)^{-1}\left(\sum_i \mathbb{E}\left[\mathbf{X}_{i,t}\mathbf{X}_{i,t}'\right]\sigma_i^2\right)\left(\sum_i \mathbb{E}\left[\mathbf{X}_{i,t}\mathbf{X}_{i,t}'\right]\right)^{-1} \tag{A.10}$$

Proof. See Supplementary Appendix S4.2. □

A.3 Bias Analysis

Consider the following model:

$$\Delta q_{i,t} = -\zeta_i \Delta p_t + u_{i,t}, \tag{A.11}$$

$$\Delta p_t = \frac{1}{\zeta_S} u_{S,t}, \tag{A.12}$$

Here, as a shorthand, we use the S subscript to denote the size-weighted sum, i.e., $\zeta_S = \sum_i S_i\zeta_i$ and $u_{S,t} = \sum_i S_i u_{i,t}$.

Importantly, the shocks $u_{i,t}$ are not orthogonal: they load on a common factor η_t, and $\epsilon_{i,t}$ is the true idiosyncratic component.

$$u_{i,t} := \lambda_i\eta_t + \epsilon_{i,t}$$

However, the econometrician fails to recognize the common factor η_t. Instead, the

[3]Note that we still use the original $\mathbf{C}_{i,t}$ in constructing the weighting without residualizing it, as $\frac{\partial\psi_{ij,t}}{\partial\zeta_k} \equiv D_{ij,k,t} = u_{jt}p_tC_{it,k} + u_{it}p_tC_{jt,k}$ regardless whether we have the common factors $X_{i,t}$ on the right-hand side.

econometrician specifies the following model:

$$\Delta q_{i,t} = -\zeta_i^b \Delta p_t + u_{i,t}^b \tag{A.13}$$

and estimates ζ_i^b using the following moment condition:

$$\mathbb{E}\left[u_{i,t}^b u_{j,t}^b\right] \equiv \mathbb{E}\left[\left(\Delta q_{i,t} + \zeta_i^b \Delta p_t\right)\left(\Delta q_{j,t} + \zeta_j^b \Delta p_t\right)\right] = 0, \tag{A.14}$$

We can show that the estimated aggregate elasticity ζ_S^b is downward biased and approximately:

$$\zeta_S^b \approx \zeta_S \sqrt{1 - \mathcal{R}^2}, \quad \mathcal{R}^2 \equiv \frac{\mathrm{Var}(\lambda_S \eta_t)}{\mathrm{Var}(u_{S,t})} = \frac{\lambda_S^2 \sigma_\eta^2}{\lambda_S^2 \sigma_\eta^2 + \sigma_{\epsilon,S}^2} \tag{A.15}$$

where $\sigma_{\epsilon,S}^2 \equiv \sum_{i=1}^N S_i^2 \sigma_{\epsilon,i}^2 \in (0, \infty)$ is the size-weighted variance of the idiosyncratic shocks, and $\mathcal{R}^2$ is the share of the hidden factor in the total variance of the (residualized) price. Equation (A.15) states that if the econometrician fails to recognize the common factor, there would be a downward bias in the estimated aggregate elasticity, and the bias is increasing in the share of the unobserved comovement.

Writing the optimal GIV as a bias-corrected OLS estimator (equation (A.7)) provides intuition for this bias formula. For an individual investor, the OLS estimator for their elasticity is downward-biased due to simultaneity—if their demand increases, this increases prices pushing demand in the opposite direction, dampening its effect. The optimal GIV corrects this bias by adjusting for the influence of their own demand shifts on price. Residual common factors lead to systematic under-correction of the downward bias, and the required correction increases with the influence of the hidden factors on price. The precise statement on the approximation is given in Proposition A.2 below, where we also further characterize the bias in individual elasticities using the individual loadings on the hidden factors and their individual contribution to the price.

To illustrate the magnitude of the bias, take the estimated aggregate elasticity $\hat{\zeta}_S^b = 1$. Equation (A.15) implies a true elasticity of 1.15 if an unrecognized common factor explains 25% of the residual variance of the demand shocks and 1.41 if it explains 50%; for the estimate to be off by 100%, the hidden factor would have to explain 75% of the residual variance. It seems unlikely to us that such a strong common factor would be unobservable and escape principal components.

Under the homogeneous case (homoskedasticity $\sigma_i = \sigma$, equal sizes $S_i = \frac{1}{N}$, and same loadings $\lambda_i = \lambda_S$), Equation (A.15) is exact. When the size distribution is unequal,

we show that heterogeneity only enters the bias formula in the second order. More precisely, we have the following proposition:

Proposition A.2. *Let the true model be given by (A.11) and (A.12). Define following deviations:*

$$\omega_i = \frac{S_i - \bar{S}}{\bar{S}}, \qquad \delta_i = \frac{\lambda_i - \lambda_S}{\lambda_S}, \qquad \kappa_i = \frac{\sigma^2_{\epsilon,i} - \bar{\sigma}^2_\epsilon}{\bar{\sigma}^2_\epsilon},$$

where $\bar{S} = \frac{1}{N}$ *and* $\bar{\sigma}^2_\epsilon = \frac{\sum_i S_i^2 \sigma^2_{\epsilon,i}}{\sum_i \bar{S}^2}$. *Let the dispersion scale* ξ *be*

$$\xi \equiv \max\left\{ \sqrt{\sum_i \omega_i^2},\ \sqrt{\sum_i S_i \delta_i^2},\ \sqrt{\sum_i S_i^2 \kappa_i^2} \right\}.$$

Write $\sigma^2_{\epsilon,S} = \sum_i S_i^2 \sigma^2_{\epsilon,i}$, $\sigma^2_{u,S} = \sigma^2_{\epsilon,S} + \lambda_S^2 \sigma^2_\eta$, $\mathcal{R}^2 = \frac{\lambda_S^2 \sigma^2_\eta}{\sigma^2_{u,S}} \in (0,1)$. *Let* ζ_S *and* ζ_i *denote the true aggregate and sectoral elasticities, and* ζ_S^b, ζ_i^b *their biased estimates.*

Then, as $\xi \to 0$,

$$\zeta_S^b = \zeta_S \sqrt{1 - \mathcal{R}^2} + O(\xi^2)$$

so dispersion affects the aggregate elasticity only at the second order. Moreover, denote $b = 1 - \sqrt{1 - \mathcal{R}^2}$. *For each sector* i,

$$\frac{\zeta_i - \zeta_i^b}{\zeta_S} = b + \mathcal{R}^2 \delta_i - b\big(1 - b\big)(\omega_i + \kappa_i) + O(\xi^2).$$

Proof. See Supplementary Appendix S4.3. □

A.4 Structural Interpretations of the Linear Demand System

In the main text, we estimate a log-linear demand system that can be interpreted as a first-order approximation of a generic portfolio choice problem. Table A.1 maps four single-asset micro-foundations to the elasticity ζ_i and the demand shifter $\nu_{i,t}$ of the log-linear demand curve. Supplementary Appendix S2.3 derives each mapping, and Appendix A.5 extends the single-asset demand curve to a multi-asset setting.

The elasticity has a different source in each model. In the CRRA benchmark, the long-run expected return and risk-free rate pin it down, and at plausible values it lies far above our estimates. Learning from prices lowers it: the more weight investors put on the current price (ϕ_p), the less they trade against price changes. With convenience services, the curvature ψ_i of utility over Treasury holdings governs it. Under liability-driven

investment, duration matching yields an elasticity below one when liabilities have longer duration than Treasuries. Wealth, risk aversion, interest rates, private signals, consumption, and liabilities enter the demand shifter $\nu_{i,t}$.

Table A.1: Mapping between Structural Parameters and Reduced-Form Elasticities

Model	Portfolio choice	Sources of ζ_i	Sources of $\nu_{i,t}$
CRRA portfolio choice (Vayanos & Vila, 2021)	$\frac{Q_{i,t}P_t}{W_{i,t}} = \frac{\mu_t - r_t}{\gamma_{i,t}\,\sigma_t^2}$	$\bar{\mu},\ \bar{r}$	$\widehat{\mathbb{E}_t[p_{t+1}]},\ \hat{w}_{i,t},\ \hat{r}_t,\ \hat{\gamma}_{i,t}$
Learning from price (Hellwig, 1980)	$\widehat{\mathbb{E}_t[p_{t+1}]} = \phi_p\,\hat{p}_t + \phi_s\,s_{i,t}$	$\phi_p,\ \bar{\mu},\ \bar{r}$	$\phi_s s_{i,t},\ \hat{w}_{i,t},\ \hat{r}_t,\ \hat{\gamma}_{i,t}$
Bond-in-utility (Krishnamurthy & Vissing-Jorgensen, 2012)	$\frac{(Q_{i,t}P_t)^{-\psi_i}}{C_{i,t}^{-\theta_i}} = r_t - y_t$	$\psi_i,\ \bar{r},\ \bar{y}$	$\frac{\theta_i}{\psi_i}\,\hat{c}_{i,t},\ -\frac{1}{\psi_i \overline{cy}}\,\hat{r}_t$
Liability-driven investment (Domanski, Shin, & Sushko, 2017)	$Q_{i,t}P_tD_t^T = L_{i,t}(y_t)D_{i,t}^L$	$\frac{\partial \log D_t^T}{\partial \log P_t},\ \frac{\partial \log D_{i,t}^L}{\partial \log P_t}$	$\hat{l}_{i,t}$

Note: Bars denote long-run levels (or log-levels when variables are in logs), and hats denote deviations from long-run levels (log-deviations when in logs). For instance, if $q_{i,t} := \log Q_{i,t}$ then $\hat{q}_{i,t} = q_{i,t} - \bar{q}_i$, while for rates we use $\hat{r}_t := r_t - \bar{r}$. $Q_{i,t}$ is sector i's Treasury holdings, P_t the Treasury market portfolio price, and $W_{i,t}$ the sector's wealth. μ_t is the one-period expected return, r_t the short risk-free rate, y_t the Treasury yield, σ_t^2 the return variance, and $\gamma_{i,t}$ risk aversion. In the bond-in-utility model, $C_{i,t}$ is consumption (not the elasticity design $\mathbf{C}_{i,t}$ of Appendix A.1), ψ_i and θ_i are the curvatures of utility over Treasury holdings and consumption, and $\overline{cy}$ is the long-run convenience yield $r - y$. $\widehat{\mathbb{E}_t[p_{t+1}]}$ is the (log) deviation of the expected future price; $s_{i,t}$ is a private signal; ϕ_p and ϕ_s are learning-from-price weights. For liability-driven investment, D_t^T and $D_{i,t}^L$ are modified durations for Treasuries and liabilities, and $L_{i,t}$ is the present value of liabilities with $\hat{l}_{i,t}$ its deviation from the long-run level.

A.5 Single-Asset Representation of Multi-Asset Demand

A multi-asset system can be collapsed to a single-asset system with a total elasticity, which embeds the general-equilibrium feedback from cross-asset substitution. This appendix formalizes this mapping and discusses why the total elasticities, rather than the partial elasticities, are typically more relevant for policy analysis.[4]

Two-Asset System. Let asset 1 be Treasuries and asset 2 be a close substitute; the logic extends naturally to an N-asset system. For exposition, assume only sector-specific demand shifters (no common shifters). Sector i's demands are:

$$q_{i,1,t} = -\zeta_{i,11}p_{1,t} + \zeta_{i,12}p_{2,t} + u_{i,1,t},$$
$$q_{i,2,t} = -\zeta_{i,22}p_{2,t} + \zeta_{i,21}p_{1,t} + u_{i,2,t},$$

[4]Our methodology yields price multipliers estimated using time-series variation. In related work, Haddad, He, Huebner, Kondor, and Loualiche (2026) discuss the interpretation of price elasticities of demand using cross-sectional instruments.

where $\zeta_{i,11}$ and $\zeta_{i,22}$ are the own-price elasticities, $\zeta_{i,12}$ and $\zeta_{i,21}$ are the cross-price elasticities, and $u_{i,1,t}$ and $u_{i,2,t}$ are the idiosyncratic demand shocks.

Treasury market impact on substitute prices. Market clearing for the substitute asset, $\sum_i S_{i,2} q_{i,2,t} = 0$, implies:

$$p_{2,t} = \underbrace{\mathcal{P}_{1\to 2}}_{:=\zeta_{S,21}/\zeta_{S,22}} p_{1,t} + \frac{1}{\zeta_{S,22}} u_{S,2,t}$$

where $\mathcal{P}_{1\to 2} \equiv \frac{\zeta_{S,21}}{\zeta_{S,22}}$ is the price pass-through coefficient from Treasuries to the substitute asset. It captures the equilibrium percentage change in the substitute asset price induced by a 1% change in Treasury prices—for a detailed discussion on pass-through coefficients see Chaudhary, Fu, and Li (2026).

Single-asset representation. Substituting p_2 into the demand for Treasuries we get,

$$q_{i,1,t} = -\underbrace{(\zeta_{i,11} - \zeta_{i,12}\mathcal{P}_{1\to 2})}_{:=\tilde{\zeta}_{i,1}} p_{1,t} + \frac{\zeta_{i,12}}{\zeta_{S,22}} u_{S,2,t} + u_{i,1,t}$$

where $\tilde{\zeta}_{i,1} \equiv \zeta_{i,11} - \zeta_{i,12}\,\mathcal{P}_{1\to 2}$ is the total elasticity, which comprises two parts. The partial elasticity $\zeta_{i,11}$ captures the direct response to Treasury price changes, holding p_2 fixed. The second term, $\zeta_{i,12}\,\mathcal{P}_{1\to 2}$, captures the indirect effect: Treasury price adjustments cause p_2 changes (by $\mathcal{P}_{1\to 2}$), which in turn induce additional Treasury demand changes due to cross-asset substitution. Intuitively, the total elasticity $\tilde{\zeta}_{i,1}$ measures the equilibrium effect of a 1% exogenous change in Treasury prices, after all prices have adjusted. Note that mapping the multi-asset system to a single-asset equation introduces an aggregate demand shifter $\frac{\zeta_{i,12}}{\zeta_{S,22}} u_{S,2,t}$ capturing allocation shifts induced by aggregate demand shocks to the substitute asset.

Macro multiplier and amplification. Applying the market clearing for asset 1, we can express the Treasury macro multiplier M_1 in terms of the aggregate total elasticity and its underlying components,

$$M_1 = \frac{1}{\tilde{\zeta}_{S,1}} = \frac{1}{\zeta_{S,11}(1 - \underbrace{\mathcal{P}_{1\to 2}\mathcal{P}_{2\to 1}}_{:=\mathcal{A}_1})}$$

where $\mathcal{P}_{2\to1} \equiv \frac{\zeta_{S,12}}{\zeta_{S,11}}$ is the price pass-through coefficient from the substitute asset to Treasuries, and $\mathcal{A}_1 \equiv \mathcal{P}_{1\to2}\mathcal{P}_{2\to1}$ is the amplification factor capturing how Treasury price changes impact substitute prices, which then feed back to impact Treasury prices. This feedback effect is geometric: a change in Treasury prices impacts substitute prices, which then feed back to impact Treasury prices, and so on. Consequently, the multiplier M_1 (which captures the total equilibrium effect on Treasury prices from a 1% change in Treasury demand) depends on: (i) the partial effect of the demand shock holding other prices fixed, $1/\zeta_{S,11}$, which is then scaled by (ii) the total amplification due to equilibrium price adjustments, given by the geometric sum $\sum_{k=0}^{\infty} \mathcal{A}_1^k = \frac{1}{1-\mathcal{A}_1}$.

Policy-relevant parameter. For most policy questions, such as the price impact of quantitative easing or of a foreign portfolio rebalancing, the object of interest is the total elasticity $\tilde{\zeta}_{i,1}$, which gives the response after substitute prices have adjusted; it is also directly estimable, whereas the full matrix of partial elasticities is high-dimensional.

B Empirical Appendix

B.1 Data

B.1.1 Holdings Data

Financial Accounts data. The main dataset we use for Treasury bonds and notes holdings comes from Table FU.210 and L.210 of the 2024Q2 Z.1 release of the Federal Reserve (Fed).[5] Our estimation sample spans 1986Q1–2023Q4. We use seasonally unadjusted transactions and holdings of Treasury notes and bonds (labelled as "other Treasury securities") whenever possible. Not all sectors in the Z.1 release separately report holdings of bills, notes and bonds. Those that do not report this breakdown are typically small holders and are lumped into the "other" category for estimation. There are, however, a number of large sectors with no information on notes and bonds holdings directly available, including the household sector (residual), pension funds and ETFs. In these cases, we assume that their holdings are entirely in Treasury notes and bonds.

Country-level foreign holdings and bank-level holdings. We use information from the Treasury International Capital system (TIC) to compile country-level holdings of long-term (bonds and notes) Treasury securities by foreign investors. We use the benchmark-

[5] https://www.federalreserve.gov/apps/fof/FOFTables.aspx

consistent estimates put together by Bertaut and Tryon (2007), Bertaut and Judson (2014), and Bertaut and Judson (2022). For July–December 2011, when the TIC methodology is in transition, we use SLT holdings and Form S transactions. From January 2012, we use the Bertaut–Judson SLT-based holdings and estimated transactions.

For individual U.S. bank holdings of Treasury notes and bonds, we use the Call Reports (FFIEC 031/041/051), downloaded from WRDS. We drop foreign branches (FFIEC 002 filers), identified with the relationship files of the NIC information system. We construct Treasury holdings separately for securities held to maturity (HTM), available for sale (AFS), and in the trading book. HTM and AFS amortized cost and fair value use consolidated values where reported and domestic-office values otherwise; trading holdings use domestic-office values.[6] Holdings are the sum of HTM, AFS, and trading fair values; our transaction proxy starts from the first difference of the sum of HTM amortized cost, AFS amortized cost, and trading fair value. After the maturity allocation described below, we subtract estimated holding gains on the lagged trading position, using CRSP clean-price returns weighted by the bank's lagged long-maturity bucket shares. To separate bills from notes and bonds, we follow Jansen, Li, and Schmid (2024): using the residual-maturity breakdown of non-trading, non-mortgage debt securities, we impute notes and bonds holdings, assuming that the bill share is stable across residual-maturity buckets and that the buckets above one year approximate notes and bonds holdings.

We retain countries with nonzero holdings of Treasury notes and bonds and nonmissing transactions and lagged holdings throughout 1986Q1–2023Q4. Bank-level data begin in 1997Q3. For each bank, we keep its longest continuous spell with positive holdings, measured both at fair value and as non-trading amortized cost plus trading fair value. The spell must have mean total assets of at least $10 billion and at least 20 usable quarters in each regime it covers, before or after 2009Q1. Remaining holdings and transactions are absorbed into residual categories labelled "all other countries" and "all other banks," respectively.

Discrepancy. Because not all sectors report Treasury notes and bonds holdings separately, and direct estimates of notes and bonds holdings and transactions are unavailable, our construction leaves valuation and coverage discrepancies. We sweep the discrepancy component into the household sector (the residual sector in the Z.1 methodology).

[6] We use domestic-office trading holdings through 2018Q1. From 2018Q2, we approximate them from consolidated holdings: 75% of JPMorgan's, reflecting its historical foreign-office share of about one quarter; zero for Citibank, whose foreign-office share approaches 100% after 2012; and the full amount for other banks, whose earlier reports show little foreign-office Treasury trading.

We reallocate seven exceptional sector-quarter flows to the household residual to prevent reporting-related exits and re-entries and small sector entries from generating mechanically extreme normalized flows in the affected sectors.[7] We set each affected sector's flow to zero and transfer the same signed amount to households, preserving aggregate transactions and leaving holdings unchanged.

Sector classification and aggregation. Table B.1 reports our aggregation of Financial Accounts sectors to categories for GMM estimation and to groups for yield decomposition.

B.1.2 Observed Common Factors

We construct ten common factors for 1986Q1–2023Q4. Four describe macro-financial conditions, four summarize survey beliefs, and two control for policy flows. Except for the futures-implied rate revision, we demean each factor and divide by its sample standard deviation over 1986Q1–2023Q4. The rate revision remains in percentage-point units. Figure B.1 reports the R^2 from regressions of Treasury price changes on these factors, added one block at a time; the full set explains more than 40% of the variation in Treasury price changes.

Uncertainty / risk sentiment (η_t^{unc}). We measure uncertainty using innovations in the logarithm of quarter-end implied equity volatility, combining the CBOE VXO and VIX indices (FRED series VXOCLS and VIXCLS). For each index, we estimate an AR(1) with an intercept on their common overlap, 1990Q1–2021Q3, and apply the estimated coefficients over that index's available history. We use VIX residuals from 1990Q2 onward and VXO residuals before then.

Inflation. We use the seasonally adjusted consumer price index for all urban consumers (FRED series CPIAUCSL), taking the observation for the last month of each quarter. Annualized quarterly inflation is $400[\log(\mathrm{CPI}_t) - \log(\mathrm{CPI}_{t-1})]$. The inflation factor is the residual from an AR(1), including an intercept, estimated over 1986Q1–2023Q4.

Short-rate expectations. We construct quarterly revisions in the futures-implied expected federal funds rate using daily settlement prices of individually dated 30-day federal funds futures contracts sourced from LSEG. At each quarter-end cutoff, we use the latest available settlement on or before that date, treating settlements more than ten cal-

[7] The affected observations are federal government defined-benefit retirement funds (1986Q2 and 1986Q3), federal government defined-contribution retirement funds (1988Q1), issuers of asset-backed securities (1999Q4, 2001Q1, and 2001Q4), and other financial business (2000Q1).

Table B.1: **Mapping from Sectors to Groups**

Sector	Category	Group
Banks in U.S.-affiliated areas	Banks	Banks and asset managers
Foreign banking offices in the U.S.	Banks	Banks and asset managers
U.S.-chartered depository institutions	Banks	Banks and asset managers
Exchange-traded funds (through 2008Q4)	Other	Households and Other
Exchange-traded funds (from 2009Q1)	ETFs	Banks and asset managers
Monetary authority	Fed	Fed
Households and nonprofit organizations	Households	Households and Other
Property-casualty insurance companies	Insurance	Banks and asset managers
Life insurance companies	Insurance	Banks and asset managers
Money market funds	Other	Households and Other
Closed-end funds	Mutual funds	Banks and asset managers
Mutual funds	Mutual funds	Banks and asset managers
Nonfinancial corporate business	Other	Households and Other
Nonfinancial noncorporate business	Other	Households and Other
State and local governments	Other	Households and Other
Government-sponsored enterprises	Other	Households and Other
Credit unions	Other	Households and Other
Other financial business	Other	Households and Other
Issuers of asset-backed securities	Other	Households and Other
Holding companies	Other	Households and Other
State and local government employee defined benefit retirement funds	Pension funds	Banks and asset managers
Federal government defined contribution retirement funds	Pension funds	Banks and asset managers
Federal government defined benefit retirement funds	Pension funds	Banks and asset managers
Private defined contribution pension funds	Pension funds	Banks and asset managers
Private defined benefit pension funds	Pension funds	Banks and asset managers
Rest of the world	Rest of world	Rest of world
Security brokers and dealers	Dealers	Banks and asset managers
Supply	Supply	Supply

Note: Italicized sectors indicate that we use data more granular than the sector-level information from the Financial Accounts. Banks and asset managers combines banks, insurers, pension funds, mutual funds, ETFs, and dealers; Households and Other combines the Households and Other categories.

endar days old as missing. For target month m and observation date d, the implied rate in percentage points is $100 - P^{\text{fut}}_{m,d}$. We combine the three contracts for a target quarter using calendar days in each month as weights. Let $\widehat{r}_{t+1|t}$ denote this implied average rate for quarter $t+1$, measured at the end of quarter t. Our factor is

$$\widehat{r}_{t+1|t} - \widehat{r}_{t+1|t-1}.$$

Both observations use contracts for the same target quarter, so the difference measures repricing of the expected rate path during quarter t. For example, the first-quarter factor compares expectations for April–June at end-March with expectations for those same

months at end-December.

Federal funds futures data are available from 1991Q1. For 1986Q1–1990Q4, we proxy the factor with the fitted value from a regression of futures revisions on quarterly changes in the effective federal funds rate, with an intercept, estimated over 1991Q1–2003Q3. The realized-rate regressor uses quarter-end observations from FRED series DFF.

Dollar. We construct an extended nominal dollar index from month-end observations of Federal Reserve trade-weighted indices. From January 1995 through December 2019, we use the broad goods index (FRED series DTWEXB); from January 2020, we extend it with monthly log changes in the broad goods-and-services index (DTWEXBGS). Before January 1995, we chain the index backward using log changes in the major-currencies goods index (DTWEXM). The early segment therefore reflects a different currency basket. We estimate an AR(1) with an intercept for the extended log index over January 1986–December 2023 and sum its three monthly residuals within each quarter to obtain the dollar innovation.

Survey factors. We use individual responses to the Philadelphia Fed's Survey of Professional Forecasters (SPF) for the current quarter's average corporate bond yield, rather than Treasury-yield forecasts whose history does not span the full sample. For forecaster j, we subtract the previous quarter's realized average Moody's Aaa yield from the current-quarter forecast:

$$b_{j,t} = \widehat{y}^{\,\mathrm{SPF}}_{j,t} - \overline{y}^{\,\mathrm{Aaa}}_{t-1}.$$

The forecast is the survey's current-quarter observation, BOND2. The realized rate is taken from the quarterly average of monthly FRED series AAA. Thus $b_{j,t}$ measures a forecasted change from the previous quarter's realization.[8] We drop missing forecasts, retain forecasters with at least four observations during 1986Q1–2023Q4, and require at least seven forecasters in each quarter. We winsorize forecast levels at the 0.5% tails before subtracting the lagged realized yield. From the resulting forecaster-by-quarter panel, we extract four common factors using HeteroPCA (Zhang, Cai, & Wu, 2022), demeaning each forecaster's series and filling missing demeaned observations with zero. The estimation includes the four extracted factors.

Lagged Treasury supply. We control for the lagged normalized Treasury supply flow. Issuance carries the negative-demand sign used in Section 3.1.

Scheduled Federal Reserve purchases. Historically, the New York Fed's Open Market

[8]Before 1990Q4, the SPF corporate-yield forecast refers to the new, high-grade corporate bond yield; from 1990Q4, it refers to Moody's Aaa yield.

Trading Desk (the Desk) used outright purchases or sales of Treasury securities as a tool to manage the supply of bank reserves to maintain conditions consistent with the Federal funds target rate set by the Federal Open Market Committee (FOMC). Hence, small Treasury market flows were present in the NY Fed's System Open Market Account (SOMA) before 2008. The nature of these flows changed drastically during the global financial crisis. From 2009 to 2024, the Fed conducted four rounds of large-scale asset purchases including Treasuries and debt of government-sponsored enterprises, commonly known as Quantitative Easings (QEs), which significantly enlarged the SOMA portfolio in long-term Treasury securities.

Once activated, the scale of the QE operations can be predicted. At the start of each round of QEs, the FOMC typically announces the total amount and the pace of the purchases to provide forward guidance to market participants. Closely following the instructions of the FOMC, the Desk at the New York Fed issues a detailed schedule for future purchases typically at a biweekly frequency. As the operations are prescheduled, they generally do not respond to the contemporaneous market development.[9]

To control for the scheduled Fed purchases, we construct a sequence of anticipated purchases by the Fed in each quarter. Outside of QE regimes, we treat all purchases as anticipated; during QE regimes, we base the anticipated purchases on the FOMC's announcements in the previous quarter. For example, on March 18, 2009, the FOMC announced it would purchase up to $300 billion of longer-term Treasury securities over the next six months. The scheduled purchase by the Fed in 2009Q2 and 2009Q3 would be $150 billion each, until further policy changes by the FOMC.

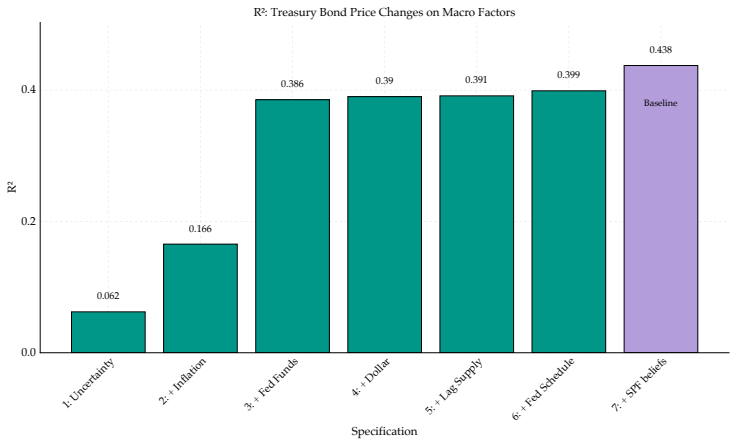
When the QE is not pre-scheduled in that quarter (typically at the start of the QEs), we use lagged purchases by the Fed from the previous quarter as the expected purchase.[10] Because investors may use more information than the FOMC announcements alone, we also treat predictable changes as anticipated. For example, during the exit from QE3, the FOMC's reductions in purchases were not pre-scheduled in earlier quarters, but because they followed a predictable pattern and were well anticipated by investors, we treat them as scheduled. Purchases outside QE periods are rollovers of maturing Treasuries or serve

[9] One exception is the initial stage of the QE in response to the COVID pandemic. Due to the extremity of the macroeconomic and financial conditions, the FOMC announced that it "will continue to purchase Treasury securities and agency mortgage-backed securities in the amounts needed to support smooth market functioning and effective transmission of monetary policy to broader financial conditions", without specifying the duration and the total amount.

[10] Alternatively, we can also estimate an AR process of the Fed purchase outside of QE and use that as predictor. The results are not sensitive to the treatment as the deviations of the QE purchase from the regular outright operations are an order of magnitude larger.

reserve management; they are small, so alternative treatments have little effect on our results. Our estimates for the Fed's sensitivity to prices and to macro factors mainly reflect the policy consideration during episodes in which Fed deviates from the anticipated schedule. Supplementary Appendix S3.1 plots the realized and anticipated paths and tabulates them during active QE periods.

Figure B.1: **Explanatory power of macro-financial factors on Treasury price changes**



Note: This figure reports the R^2 from regressions of Treasury price changes on the observed factors of Appendix B.1.2, added in the order shown. The purple bar reports the R^2 using the set of factors included in our estimation.

B.2 Robustness of Identification

This subsection reports the checks on the identification strategy summarized in Section 3.4. We first drop one sector at a time from the moment conditions, then replace the survey-based factors with internally estimated principal components, then examine the correlations of the estimated idiosyncratic shocks across sectors, and finally check the shocks against narrative evidence and the term structure.

Leave-one-out estimates. Our identification strategy uses other sectors' shocks to instrument for the price in each sector's demand, so it relies on the shocks being idiosyncratic and independent across sectors. Section 3.4 discusses how correlated shocks could arise, for example from sophisticated investors who predict and front-run other sectors' demand shocks. We therefore re-estimate the model leaving out one sector at a time. Leaving out a sector reduces statistical power, but if the identification assumption holds, the estimated elasticities should remain stable. Figures B.2 and B.3 present time-average price elasticities and post-minus-pre-2009 changes from the specification with a 2009Q1

regime shift in every parameter; Supplementary Appendix S3.4 tabulates the estimates. The leave-one-out average aggregate price elasticity ranges from 0.97 to 1.04, very close to the full-fit average of 1.01, and its decline across 2009Q1 keeps its sign and size in every leave-one-out specification. The decline in the rest of the world's elasticity is also stable across the leave-one-out specifications. These results suggest that correlation between any single sector's demand shocks and those of the other sectors, whether from strategic behavior or other sources, does not drive the estimates.

Internal principal components and bias formula diagnostics. To allow for unobserved common factors, we replace the moment conditions (A.2) with

$$\mathbb{E}\left[\hat{u}_{i,t}(\boldsymbol{\zeta})\,\hat{u}_{j,t}(\boldsymbol{\zeta})\right] = \hat{\boldsymbol{\lambda}}_i(\boldsymbol{\zeta})'\hat{\boldsymbol{\lambda}}_j(\boldsymbol{\zeta}), \qquad i \neq j, \tag{B.1}$$

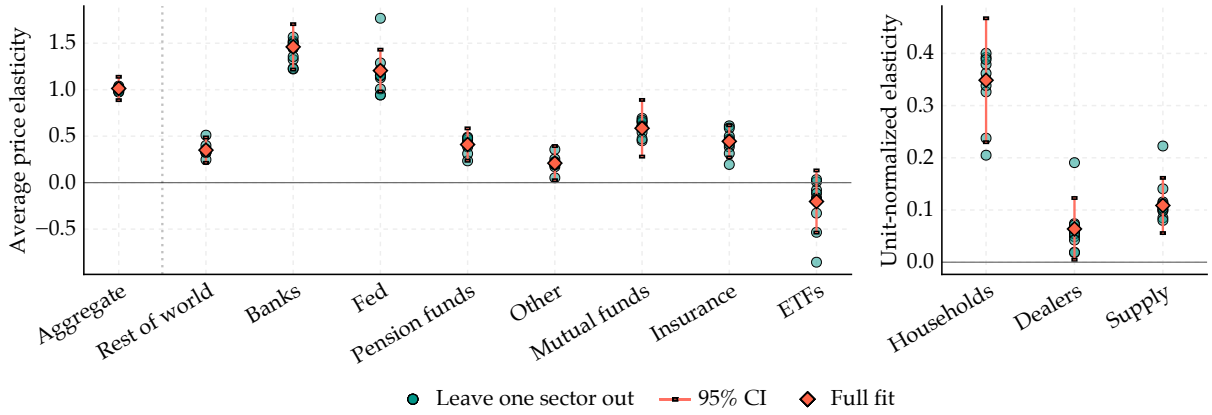
where the loadings come from a heteroskedasticity-robust PCA of the residuals at each trial $\boldsymbol{\zeta}$ (Supplementary Appendix S2.1). Figure B.4 shows that our key takeaways remain robust after replacing the survey-based common factors with these internally estimated principal components of flows (see the discussion in Section 3.4). Panel (a) plots our baseline estimates against estimates obtained with one to three internal principal components in place of the survey factors. Panel (c) reports the R^2 and adjusted R^2 obtained by projecting Treasury price changes onto the estimated internal principal components. The maximum R^2 is 2.8% with 3 internal PCs, and the adjusted R^2 is 0.9%. Guided by the approximate bias formula (3.5; Appendix A.3), we conclude that the bias arising from our baseline specification is small.

Correlations of idiosyncratic shocks $\hat{u}_{i,t}$. Pairwise sample correlations of the extracted demand and supply shocks, at the sector level ($\hat{u}_{i,t}$), are in general close to zero, including when the rest-of-world residual is split into China, Japan, and Europe (Figures B.5a and B.5b).

China as a case study. The estimated sector-level demand shocks $\hat{u}_{i,t}$ can be checked against narrative evidence. Figure B.6 shows that the extracted demand shocks for China are well aligned with narrative evidence on the role of China's domestic factors in influencing its Treasury demand.

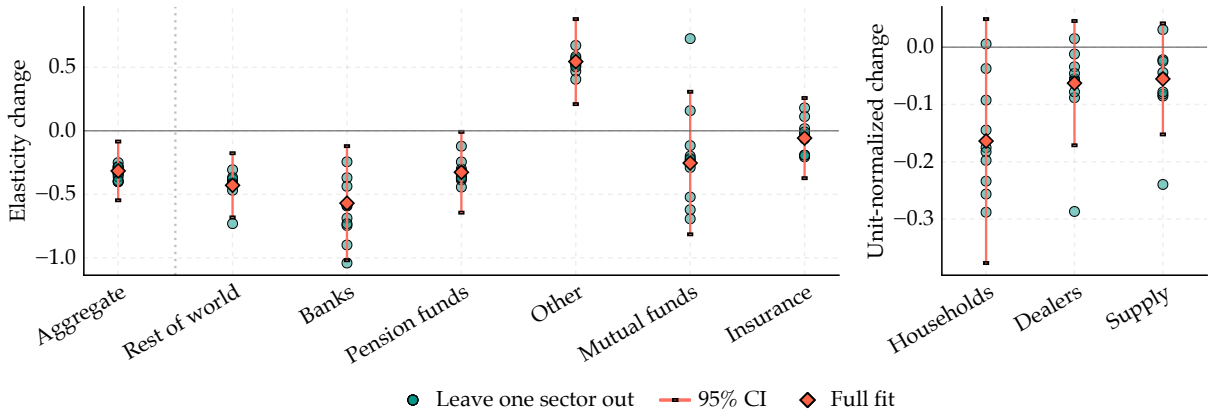
The price impact across the term structure. Our procedure yields a macro multiplier for the market portfolio of Treasury notes and bonds, as well as a set of aggregate and sectoral demand and supply shocks. Following the literature on the factor structure of yields

Figure B.2: **Price elasticity by sector: Leave-one-out estimates**



Note: This figure plots average price elasticities from leave-one-out estimates. Each leave-one-out estimate re-estimates the specification of Table 1 without the moment conditions of one sector, and each sector's elasticity is averaged over its observed quarters. The averages for the Fed and ETFs cover 2009Q1–2023Q4. Red diamonds are the estimates using all sectors, with 95% confidence intervals from equation (A.5); teal circles are the estimates after leaving out one other sector. The aggregate elasticity combines the remaining sectors' re-estimated contributions with the left-out sector's contribution from the estimation using all sectors. Households, dealers, and supply have unit size weights and are shown in the right panel.

Figure B.3: **Change in price elasticities across 2009Q1: Leave-one-out estimates**



Note: This figure plots the post- minus pre-2009 change in price elasticities from the leave-one-out estimates of Figure B.2. Marker conventions and the construction of aggregate elasticity follow Figure B.2. The Fed is omitted because its elasticity is not estimated before 2009 and enters as zero, so its change is just its post-2009 elasticity. Households, dealers, and supply are shown in the right panel.

(Litterman & Scheinkman, 1991, among others), in Figure B.7 we regress price changes implied in Treasury zero-coupon yields (Gurkaynak, Sack, & Wright, 2007) and term premia (Adrian, Crump, & Moench, 2013) across various tenors on the estimated aggregate demand and supply shocks $\hat{u}_{S,t}$. The price impact estimates increase with maturity and line up closely with a duration-based extrapolation of the price impact on the market portfolio that assumes a uniform impact on yields across maturities. This result suggests that our demand shocks mainly shift the level of the yield curve.

B.3 Additional Results

B.3.1 Multipliers Across the Literature

We compare our Treasury macro multiplier estimates with those from the literature. The focus on estimating multiplier parameters—the percentage change in prices to a percentage change in demand as a share of outstanding Treasury securities—is a recent development. However, the literature has long studied the price impact of demand shocks arising from foreign FX interventions, quantitative easing, Treasury repurchase operations, and other policy interventions. By making a few assumptions about the implicit size of the quantity changes, we can approximately map these price impact estimates from the literature to multipliers.

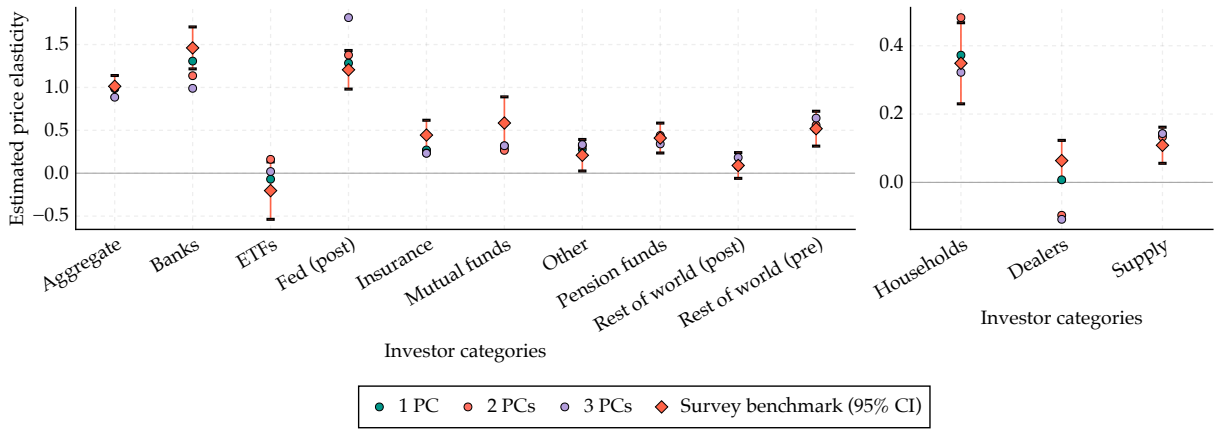
Figure B.8 plots the comparison. The y-axis shows the macro multiplier—defined as the percentage change in prices relative to the percentage change in demand as a share of outstanding Treasury securities. The x-axis groups estimates by the different types of demand shocks and methodologies they used to measure price impact. These include foreign flow shocks, quantitative easing (QE) and supply shocks, and structural methods that estimate Vayanos and Vila (2021)-style models to derive macro multipliers. With the exception of a couple of outliers, our estimate of around 0.94 is in line with the macro multiplier estimates implied by the literature. Supplementary Appendix S3.3 details the mapping for each study.

B.3.2 Yield Decomposition by Sector

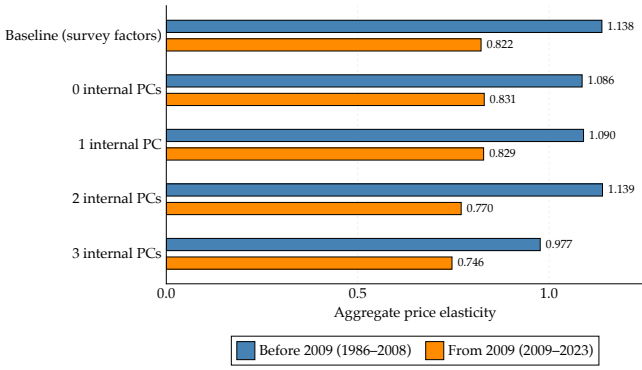
Figure B.9 reports the decomposition in equation (2.4) of the main text by group, averaged within seven episodes, using the regional elasticity specification in Table 2, panel (b).

Figure B.4: **Robustness: Internal principal components in place of the survey factors**

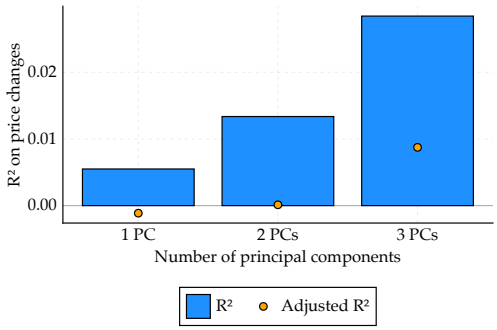
(a) Estimated price elasticities



(b) Average aggregate price elasticity before and after 2009



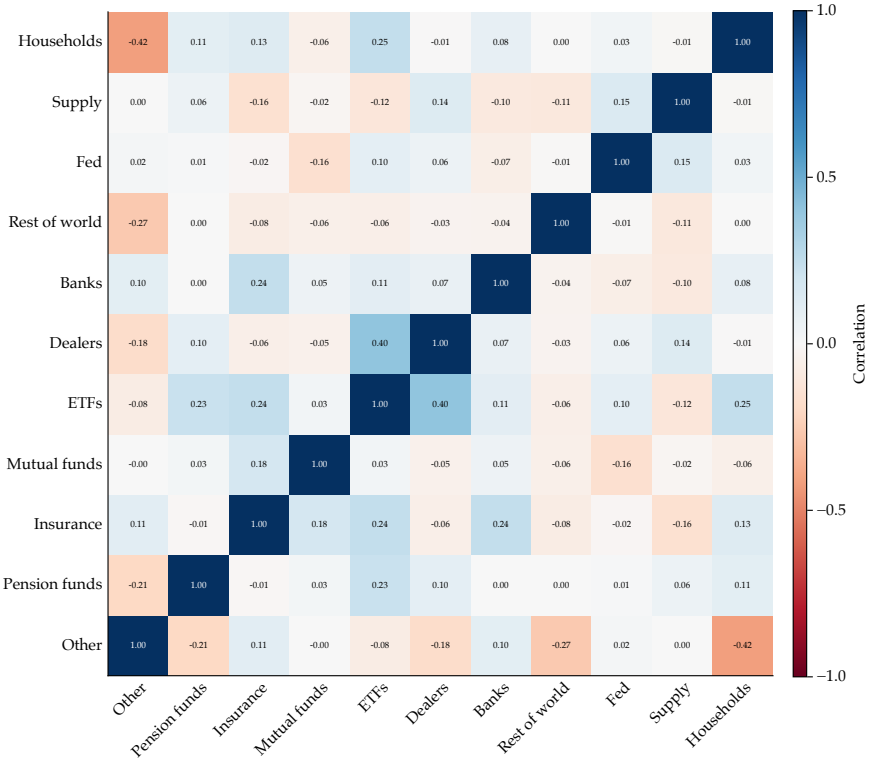
(c) Total price variation explained



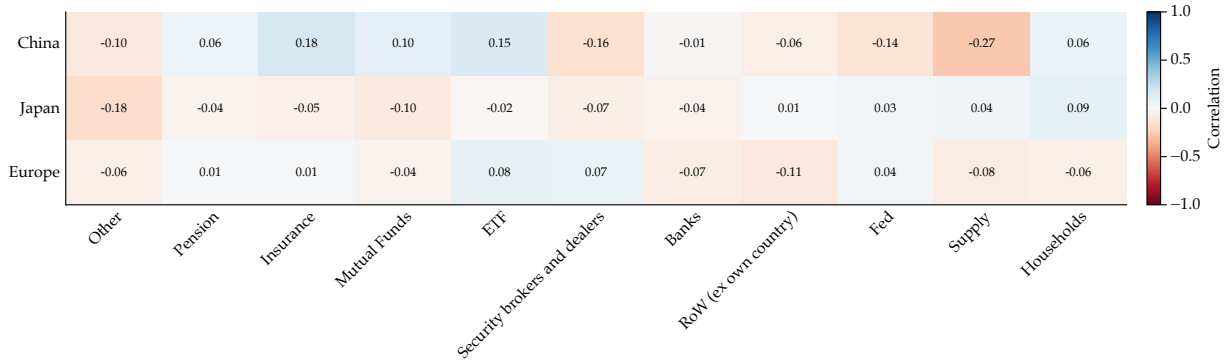
Note: This figure replaces the SPF survey-forecast factors with one to three internal principal components, estimated jointly with the model from flows over 1986Q1–2023Q4. Elasticities, intercepts, and loadings on the six other observed factors change in 2009Q1, as in Table 1; the loadings on the principal components do not. Panel (a) plots time-average sector elasticities. Red diamonds, with 95% confidence intervals from equation (A.5), are the estimates with survey factors; small diamonds are point estimates with principal components. Rest of world is shown separately before and after 2009, and the Fed after 2009 only. Households, dealers, and supply, which have unit size weights, are shown in the right panel. Panel (b) plots the average aggregate price elasticity over 1986Q1–2008Q4 and 2009Q1–2023Q4 with survey factors, without them, and with one to three principal components in their place; the specifications are otherwise as in panel (a). Each quarter's aggregate elasticity uses that quarter's size weights, and quarters are weighted equally. Panel (c) reports the full-sample R^2 and adjusted R^2 from regressing price changes on the principal components alone, with an intercept.

Figure B.5: **Correlations of estimated demand shocks across sectors**

(a) All sectors

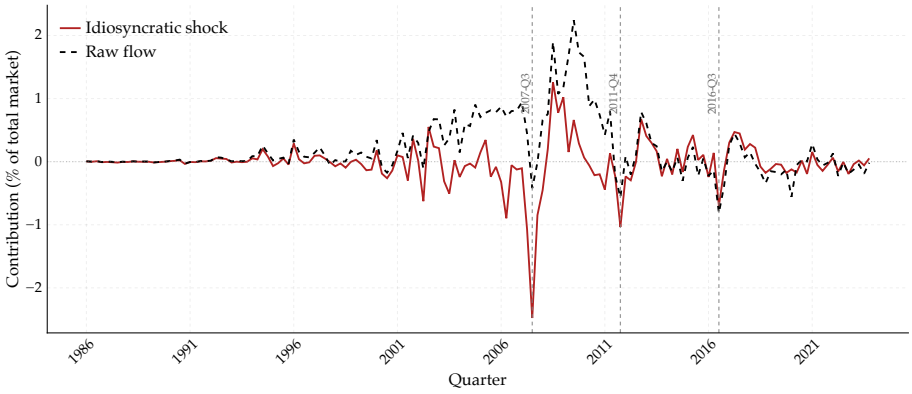


(b) China, Japan, and Europe versus other investor types



Note: Panel (a) plots pairwise sample correlations of the estimated idiosyncratic demand and supply shocks across sectors. Panel (b) plots sample correlations of the size-weighted idiosyncratic demand shocks of China, Japan, and Europe with those of other investor types over 1986Q1–2023Q4, using the quarters in which both are observed. In each row, the rest-of-world column excludes that row's region.

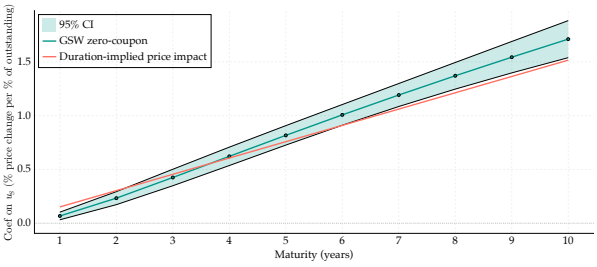
Figure B.6: **Interpreting estimated demand shocks: China as a case study**



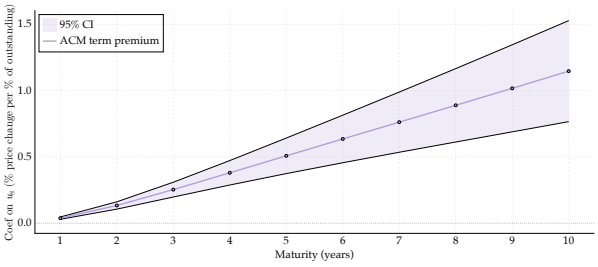
Note: This figure plots China's observed Treasury bond purchases and the idiosyncratic demand shocks estimated using optimal GIV. The large shocks line up with narrative evidence on domestic drivers of China's Treasury demand that are unrelated to global financial conditions or U.S. developments: the slowdown in China's purchases of U.S. securities in 2007Q3, read by analysts as the start of reserve diversification (USCC Economic Issue Brief); the slowdown in reserve accumulation in 2011Q4 as the RMB traded near the weak side of its band (U.S. Treasury); the sales to defend the RMB in 2016Q3 (Bloomberg); and the purchases of 2017 that stemmed appreciation pressure (Reuters).

Figure B.7: **Price impact of estimated demand shocks across maturity**

(a) Zero-coupon yields

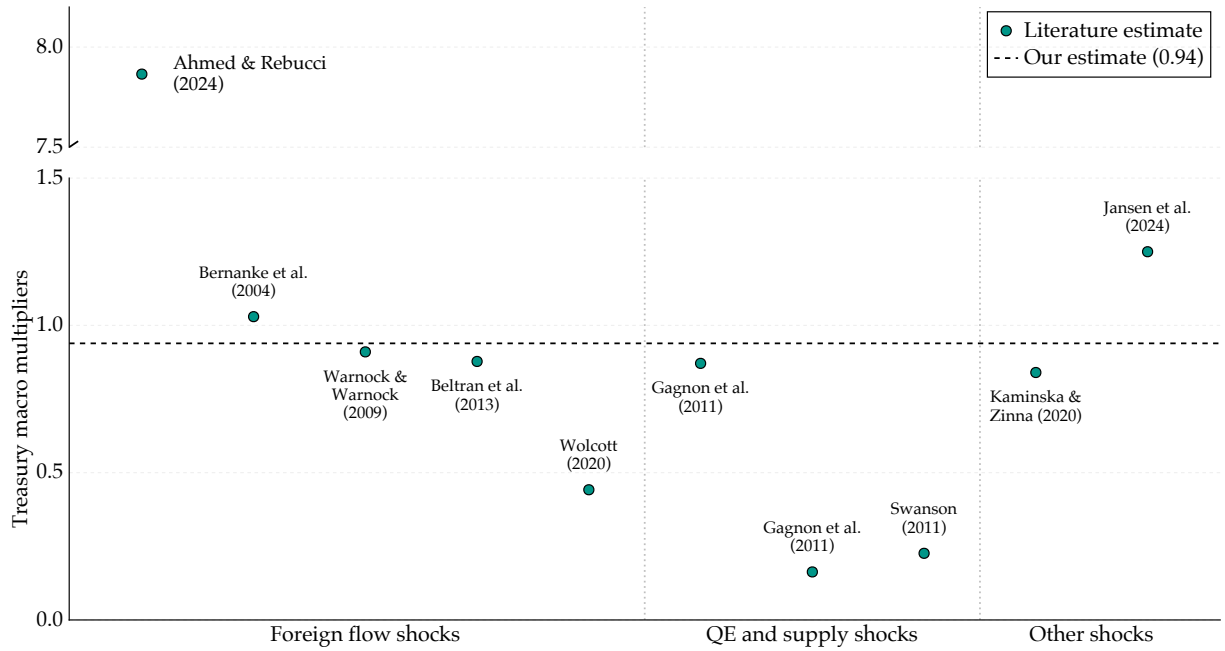


(b) Term premia



Note: This figure reports coefficients and heteroskedasticity-robust 95% confidence intervals from regressions of maturity-specific price changes on the estimated aggregate demand and supply shocks $\hat{u}_{S,t}$. Price changes are implied by zero-coupon yields (Gurkaynak, Sack, & Wright, 2007) in panel (a) and by term-premium estimates (Adrian, Crump, & Moench, 2013) in panel (b). The solid line in panel (a) is the price impact of a yield change that is uniform across maturities, scaled to match the estimated price impact on the market portfolio of notes and bonds at its average duration.

Figure B.8: **Comparing Multiplier Estimates Across the Literature**



Note: This figure compares our Treasury macro multiplier estimates with those from the literature. When a paper directly estimates macro multipliers, we present them. When a paper only estimates price impact, we map it to a multiplier by making assumptions about the implied quantity change. Our benchmark is the macro multiplier from a single-regime specification, in which private-sector price elasticities and factor loadings are constant over the sample, computed as the inverse of full-sample average aggregate price elasticity over 1986Q1–2023Q4. The vertical-axis break omits multipliers between 1.5 and 7.5.

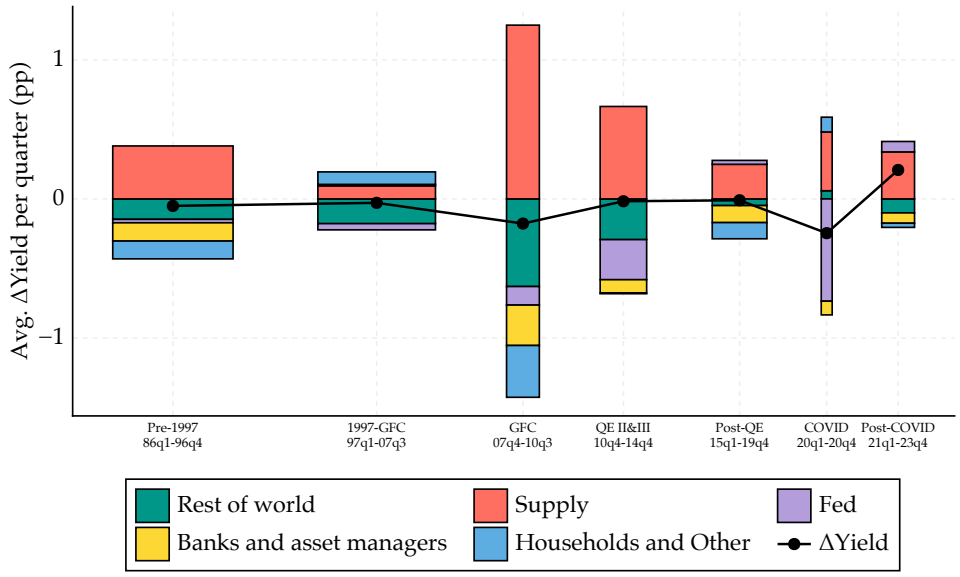
B.3.3 Safe Haven: Direct Loadings, Raw Flows, and Timing

Total and direct loadings. In equation (3.1) of the main text, sector i's demand loads on the standardized uncertainty innovation η_t^{unc} with the regime-specific coefficient $\lambda_{i,r}^{unc}$ and on the other seven contemporaneous factors of Appendix B.1.2, η_t^k for $k = 1, \ldots, 7$, with coefficients λ_i^k pooled across regimes. The *direct* loading $\lambda_{i,r}^{unc}$ is the response to a one-standard-deviation uncertainty innovation holding the other factors fixed. The other factors move with uncertainty, however; when uncertainty rises, for example, investors revise down the expected path of the federal funds rate. The *total* loading adds the responses that run through these co-movements,

$$\lambda_{i,r}^{unc,\text{total}} = \lambda_{i,r}^{unc} + \sum_{k=1}^{7} b_{k,r}\, \lambda_i^k,$$

where $b_{k,r}$ is the regime-specific slope from a regression of η_t^k on η_t^{unc}, lagged Treasury supply, and scheduled Federal Reserve purchases. The last two are predetermined, so an

Figure B.9: **Contribution to Treasury yield fluctuations by investor and issuer groups**



Note: This figure plots each group's average quarterly contribution to Treasury yield changes within seven episodes over 1986Q1–2023Q4, with bar widths reflecting episode lengths; GFC denotes the global financial crisis. Each group's contribution is the size-weighted sum of its entities' factor responses, intercepts, and idiosyncratic shocks, divided by that quarter's aggregate price elasticity and converted to a yield change at the lagged Treasury duration. The five contributions sum to the observed yield change. Table B.1 maps sectors to groups.

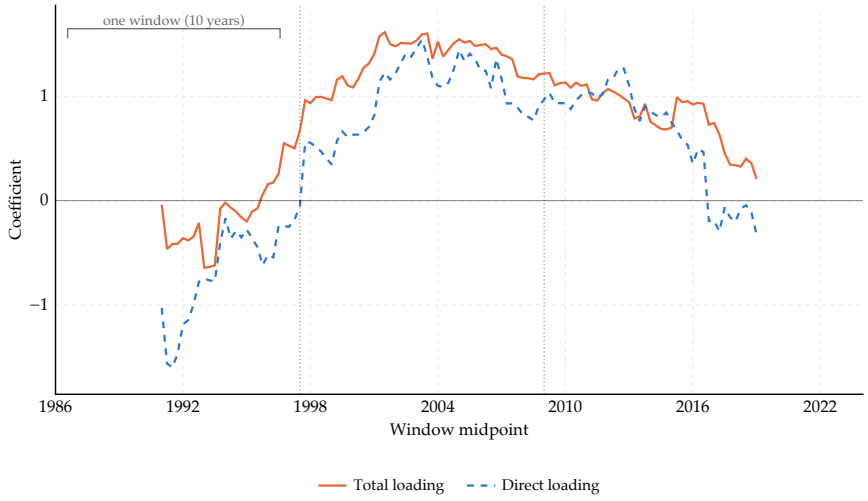
uncertainty innovation cannot move them; they are held fixed in both loadings. Standard errors of total loadings add a heteroskedasticity-robust variance of $\hat{b}_{k,r}$ to the loading variance (A.9); the two are asymptotically uncorrelated because $u_{i,t}$ is mean-independent of the factors (Assumption A.1).

Direct-loading counterparts. Figure B.10 plots the ten-year rolling price regression with both loadings, and Table B.2 repeats Table 3 with direct loadings. The three-regime conclusions about dealers and foreign investors are the same: dealers dominate both transitions, and foreign investors make no positive contribution. Supplementary Appendix S3.5 gives the direct-loading versions of Figures 4 and 5 (Figures S3.4 and S3.5) and reports each sector's price elasticity and its direct and total uncertainty responses in the first and third regimes.

Raw flows. Observed flows show the same patterns before any estimation. Scaled by each group's lagged holdings and net of the growth of privately held debt, foreign flows fall and domestic private flows rise with uncertainty, both over the full sample and during

the crisis (Figure B.11). Dealers' net purchases rise with uncertainty only in the middle regime (Figure B.12).

Figure B.10: **Ten-year rolling price loading on uncertainty innovations**

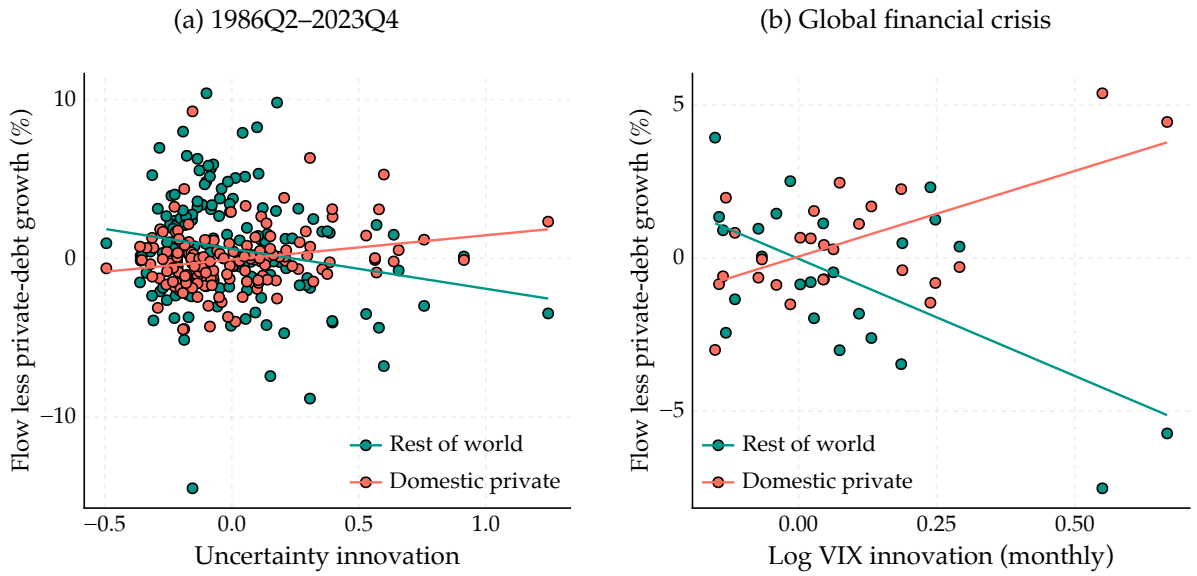


Note: This figure reports 10-year rolling OLS regressions of the quarterly price change of Treasury notes and bonds on the uncertainty innovation. The solid line is the total loading, from the univariate regression; the dashed line is the direct loading, controlling for the seven other contemporaneous factors and the two policy controls. Windows are plotted at their midpoints; dotted vertical lines mark the regime shifts at 1997Q3 and 2009Q1.

Timing of the first regime shift. Section 5 places the first regime shift at 1997Q3. We use the price regression to locate the window in which the safe-haven property emerged and examine how the demand-system results change as the first regime shift moves within that window. Figure B.13a re-estimates the price regression of Figure B.10 with two regime shifts, holding the second at 2009Q1 and moving the first from 1990Q1 to 2004Q4. The fit is best when the first shift falls in 1996–1998: the sum of squared residuals stays within 1% of its minimum over 1996Q2–1998Q3 for the total loading and over 1996Q2–1997Q3 for the direct loading. The rise in the price loading is also largest there; at 1997Q3 it is 1.65 percentage points for the total loading and 1.53 for the direct loading. Within this window, we place the shift at 1997Q3, when our bank-level flows in Treasury notes and bonds first become available. The Call Report maturity breakdown used to separate notes and bonds from bills is available from 1997Q2, and our flow measure requires two consecutive quarters.

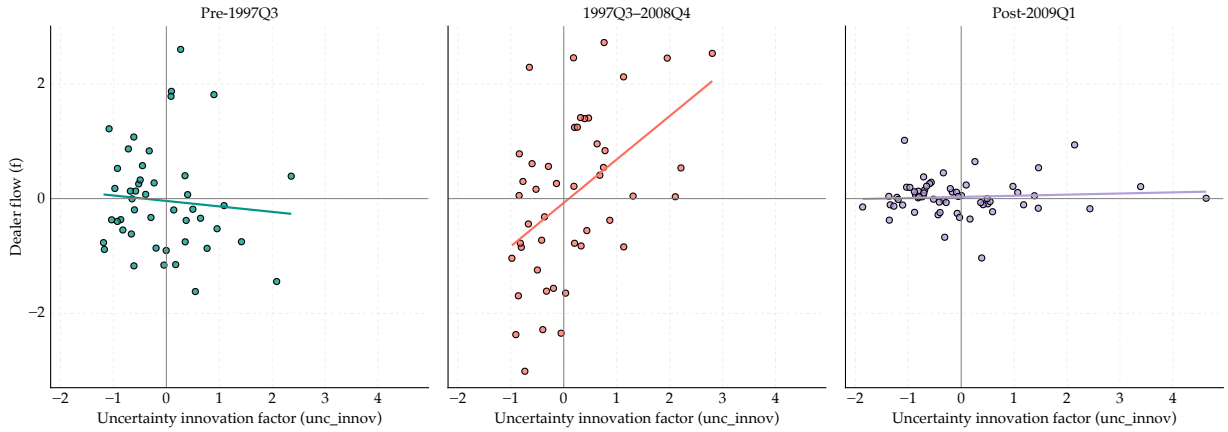
The left and center panels of Figure B.13b re-estimate the three-regime demand system of Table 3 with the first regime shift at each quarter from 1995Q1 to 1999Q4 and

Figure B.11: **Foreign and domestic private flows (net of private-debt growth) and uncertainty**



Note: This figure plots foreign and domestic private net flows into Treasury securities against uncertainty innovations. Domestic private holdings and flows equal privately held gross federal debt (Dallas Fed) less foreign Treasury holdings and flows. Each group's flow is divided by its own lagged holdings, and the growth rate of privately held debt is then subtracted. Panel (a) uses quarterly innovations in log implied volatility from 1986Q2 to 2023Q4; panel (b) uses monthly innovations in log VIX from December 2007 to December 2009.

Figure B.12: **Dealer flows and uncertainty innovations, by regime**



Note: This figure plots the quarterly dealer net flows used in the demand estimation, normalized by total Treasuries outstanding, against the standardized uncertainty innovation, 1986Q1–2023Q4, with one panel per regime. Lines are OLS fits.

Table B.2: **Three-regime direct uncertainty responses**

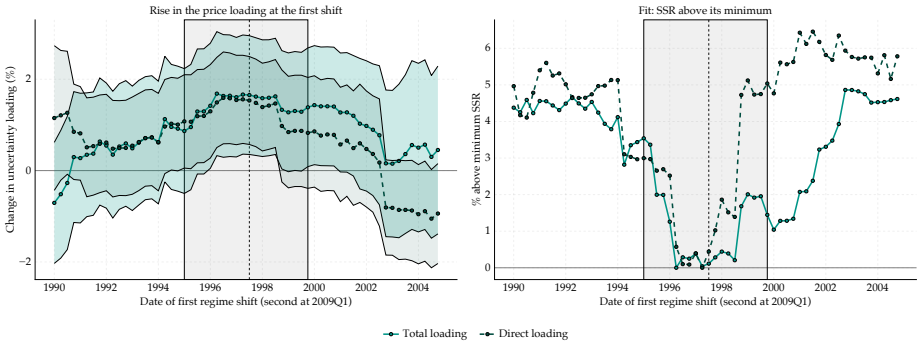
	1986Q1–1997Q2		1997Q3–2008Q4		2009Q1–2023Q4	
	Loading	Contribution	Loading	Contribution	Loading	Contribution
Aggregate	-1.12^{**}	—	1.65^{***}	—	0.11	—
	[−2.20, −0.03]		[0.56, 2.73]		[−0.65, 0.88]	
(a) Contributions to the aggregate						
Rest of world	−0.16	−0.03	0.00	0.00	−0.61	−0.29
	[−1.56, 1.24]	[−0.32, 0.25]	[−1.37, 1.36]	[−0.64, 0.63]	[−1.66, 0.43]	[−0.77, 0.20]
Households†	−0.15	−0.15	0.72^{**}	0.72^{**}	0.42^{**}	0.42^{**}
	[−0.77, 0.46]	[−0.77, 0.46]	[0.09, 1.34]	[0.09, 1.34]	[0.05, 0.78]	[0.05, 0.78]
Dealers†	−0.13	−0.13	1.07^{***}	1.07^{***}	−0.07	−0.07
	[−0.52, 0.26]	[−0.52, 0.26]	[0.68, 1.47]	[0.68, 1.47]	[−0.31, 0.17]	[−0.31, 0.17]
Remaining domestic	-0.89^{*}	-0.61^{*}	−0.41	−0.18	−0.50	−0.16
	[−1.89, 0.12]	[−1.30, 0.08]	[−1.43, 0.60]	[−0.61, 0.26]	[−1.20, 0.20]	[−0.39, 0.07]
Supply†	−0.19	−0.19	0.04	0.04	0.04	0.04
	[−0.55, 0.18]	[−0.55, 0.18]	[−0.33, 0.40]	[−0.33, 0.40]	[−0.19, 0.26]	[−0.19, 0.26]
Fed					0.83	0.18
					[−0.27, 1.94]	[−0.06, 0.42]
(b) Rest of world vs. domestic private						
Domestic private	-1.19^{*}	-0.90^{*}	3.12^{***}	1.61^{***}	0.49	0.18
	[−2.51, 0.13]	[−1.89, 0.10]	[1.50, 4.75]	[0.77, 2.45]	[−0.83, 1.80]	[−0.31, 0.67]
Difference	1.03	0.86	-3.13^{***}	-1.61^{***}	−1.10	−0.47
	[−0.90, 2.96]	[−0.17, 1.90]	[−5.29, −0.97]	[−2.68, −0.54]	[−2.78, 0.59]	[−1.16, 0.23]

Note: This table repeats Table 3 with direct uncertainty loadings, which hold the other factors fixed, in place of total loadings. Rows, contributions, and the specification are as in Table 3. 95% confidence intervals, from equation (A.9), are in brackets below estimates. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

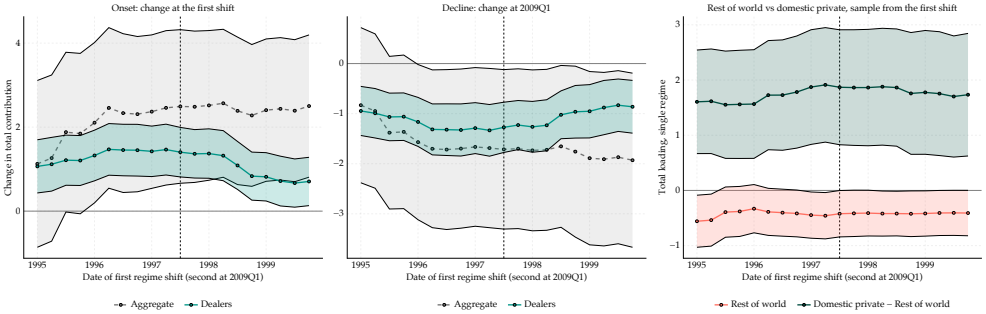
the second fixed at 2009Q1. The aggregate loading rises significantly at the first shift at 16 of the 20 dates; dealers' contribution rises significantly at every date through 1998Q4 and falls significantly at 2009Q1 in all 20 specifications. The right panel re-estimates the loadings on samples that begin at each candidate date and end in 2023Q4. The rest of the world's total loading is negative at every date, and domestic private investors' total loading exceeds it at every date by a margin significant at 5%.

Figure B.13: **Safe-haven estimates with alternative dates for the first regime shift**

(a) Treasury price regressions



(b) Demand-system uncertainty loadings



Note: Panel (a) re-estimates the price regression of Figure B.10 over 1986Q1–2023Q4, with regime shifts in slope and intercept at the first regime shift and at 2009Q1. The total loading comes from the univariate regression on uncertainty; the direct loading controls for the seven other contemporaneous factors and the two policy controls. Left: rise in the loading at the first shift. Right: sum of squared residuals relative to its minimum. The shaded span marks the first-shift dates of panel (b). Panel (b), left and center: changes in dealers' contribution and in the aggregate total loading at the first shift (left) and at 2009Q1 (center), from the specification of Table 3 with the first shift at the quarter shown. Panel (b), right: total loadings estimated on a single regime from the quarter shown to 2023Q4, with a 2009Q1 regime shift in foreign price elasticities; banks are aggregated because detailed bank data start in 1997Q3. Bands are 95% confidence intervals, heteroskedasticity-robust in panel (a) and computed as in Table 3 in panel (b); dashed lines mark 1997Q3.

References

Adrian, T., Crump, R. K., & Moench, E. (2013). Pricing the term structure with linear regressions. *Journal of Financial Economics*, *110*(1), 110–138.

Bertaut, C. C., & Judson, R. (2022). *Estimating U.S. Cross-Border Securities Flows: Ten Years of the TIC SLT*. Board of Governors of the Federal Reserve System (U.S.) Retrieved from https://ideas.repec.org/p/fip/fedgfn/2022-02-18-2.html

Bertaut, C. C., & Judson, R. A. (2014). *Estimating U.S. Cross-Border Securities Positions: New Data and New Methods* (No. 1113). Board of Governors of the Federal Reserve System (U.S.) Retrieved from https://ideas.repec.org/p/fip/fedgif/1113.html

Bertaut, C. C., & Tryon, R. W. (2007). *Monthly estimates of U.S. cross-border securities positions* (No. 910). Board of Governors of the Federal Reserve System (U.S.) Retrieved from https://ideas.repec.org/p/fip/fedgif/910.html

Chaudhary, M., Fu, Z., & Li, J. (2026). Corporate Bond Multipliers: Substitutes Matter. *The Review of Financial Studies*.

Domanski, D., Shin, H. S., & Sushko, V. (2017). The Hunt for Duration: Not Waving but Drowning? *IMF Economic Review*, *65*(1), 113–153.

Gabaix, X., & Koijen, R. S. J. (2024). Granular Instrumental Variables. *Journal of Political Economy*, *132*(7), 2274–2303.

Gurkaynak, R. S., Sack, B., & Wright, J. H. (2007). The U.S. Treasury yield curve: 1961 to the present. *Journal of Monetary Economics*, *54*(8), 2291–2304.

Haddad, V., He, Z., Huebner, P., Kondor, P., & Loualiche, E. (2026, February 27). Causal Inference for Asset Pricing.

Hellwig, M. F. (1980). On the aggregation of information in competitive markets. *Journal of Economic Theory*, *22*(3), 477–498.

Jansen, K. A. E., Li, W., & Schmid, L. (2024). *Granular Treasury Demand with Arbitrageurs*.

Krishnamurthy, A., & Vissing-Jorgensen, A. (2012). The Aggregate Demand for Treasury Debt. *Journal of Political Economy*, *120*(2), 233–267.

Litterman, R. B., & Scheinkman, J. (1991). Common Factors Affecting Bond Returns. *The Journal of Fixed Income*, *1*(1), 54–61.

Vayanos, D., & Vila, J.-L. (2021). A Preferred-Habitat Model of the Term Structure of Interest Rates. *Econometrica*, *89*(1), 77–112.

Zhang, A. R., Cai, T. T., & Wu, Y. (2022). Heteroskedastic PCA: Algorithm, optimality, and applications. *The Annals of Statistics*, *50*(1), 53–80.